

Stochastic Differential Equations

math-foundations / 33-sde

First, an example

Before the abstract framework, consider the *Ornstein–Uhlenbeck* (OU) process $dX_t = -\theta X_t dt + \sigma dW_t$ with $\theta = 1$, $\sigma = 0.5$, $X_0 = 2$. The drift $-\theta X_t$ pulls the state back toward 0; the noise term σdW_t kicks it around with increments scaling like $\sqrt{dt}$ (not dt , the key Brownian scaling). Discretely, the Euler–Maruyama update is $X_{n+1} = X_n - \theta X_n \Delta t + \sigma \sqrt{\Delta t} Z_n$ with $Z_n \sim \mathcal{N}(0, 1)$ iid. Starting at $X_0 = 2$, the mean decays like $2e^{-t}$ while the variance grows toward the stationary value $\sigma^2/(2\theta) = 0.25/2 = 0.125$ as $t \rightarrow \infty$. Keep this picture in mind: deterministic pull + $\sqrt{dt}$ -scaled noise.

1 From ODE to SDE

Classical ODEs $\dot{x} = b(x, t)$ describe smooth deterministic dynamics. To model systems perturbed by “continuous” noise, replace the increment $dx = b(x, t) dt$ with

$$dX_t = b(X_t, t) dt + \sigma(X_t, t) dW_t,$$

Read aloud: “dee-X-sub-t equals b of X-sub-t comma t, dee-t, plus sigma of X-sub-t comma t, dee-W-sub-t.” where W_t is a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$. Because Brownian paths have unbounded total variation, this expression is shorthand for the Itô integral equation

$$X_t = X_0 + \int_0^t b(X_s, s) ds + \int_0^t \sigma(X_s, s) dW_s.$$

Read aloud: “X-sub-t equals X-naught plus the integral from zero to t of b of X-sub-s comma s, dee-s, plus the Itô integral from zero to t of sigma of X-sub-s comma s, dee-W-sub-s.”

Definition 1 (SDE, drift, diffusion). *A (one-dimensional, time-inhomogeneous) Itô SDE is the relation $dX_t = b(X_t, t) dt + \sigma(X_t, t) dW_t$ with X_0 an $\mathcal{F}_0$ -measurable random variable. The map $b : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ is the drift coefficient; $\sigma : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ is the diffusion coefficient. A strong solution is an adapted continuous process X_t satisfying the integral equation a.s.*

Remark 1 (Existence/uniqueness). *If b, σ are uniformly Lipschitz in x and of linear growth, then for any $X_0 \in L^2$ independent of W there exists a unique strong solution with $\mathbb{E} \sup_{t \leq T} |X_t|^2 < \infty$. The proof is Picard iteration on the Banach space $L^2(\Omega; C([0, T]))$ with the Itô isometry replacing $|\cdot|$ estimates in the stochastic integral.*

2 Discretization: Euler–Maruyama

Definition 2 (Euler–Maruyama scheme). *Fix $N \in \mathbb{N}$, $\Delta t = T/N$, $t_n = n \Delta t$. The Euler–Maruyama (EM) discretization of $dX_t = b(X_t, t) dt + \sigma(X_t, t) dW_t$ is*

$$X_{n+1} = X_n + b(X_n, t_n) \Delta t + \sigma(X_n, t_n) \sqrt{\Delta t} Z_n, \quad Z_n \stackrel{iid}{\sim} \mathcal{N}(0, 1).$$

Read aloud: “ X -sub- n -plus-one equals X -sub- n plus b of X -sub- n comma t -sub- n , times Δt , plus sigma of X -sub- n comma t -sub- n , times square-root Δt , times Z -sub- n ; where Z -sub- n is iid standard normal.”

Theorem 1 (Strong convergence of Euler–Maruyama). *Let $b, \sigma : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ be uniformly Lipschitz in x , $\frac{1}{2}$ -Hölder in t , and of linear growth. Let X_t be the strong solution of $dX_t = b(X_t, t) dt + \sigma(X_t, t) dW_t$ on $[0, T]$ with $X_0 \in L^2$, and let $\{X_n\}_{n=0}^N$ be the EM approximation on the same Brownian path with step $\Delta t = T/N$. Then there exists a constant $C = C(T, K, \mathbb{E}|X_0|^2)$ such that*

$$\mathbb{E}[|X_{t_n} - X_n|] \leq C \sqrt{\Delta t} \quad \text{for all } 0 \leq n \leq N.$$

Read aloud: “the expectation of the absolute value of X -at- t -sub- n minus X -sub- n is at most C times square-root Δt , for all n between zero and N .” That is, EM has strong order $1/2$. (Its weak order is 1.)

Heuristic sketch. Fix a step $[t_n, t_{n+1}]$ and write $\Delta W_n := W_{t_{n+1}} - W_{t_n}$, so $\Delta W_n \sim \mathcal{N}(0, \Delta t)$ and $\mathbb{E}[(\Delta W_n)^2] = \Delta t$. Taylor-expand the exact increment of X :

$$X_{t_{n+1}} - X_{t_n} = \underbrace{b(X_{t_n}, t_n) \Delta t}_{A_n} + \underbrace{\sigma(X_{t_n}, t_n) \Delta W_n}_{B_n} + R_n,$$

Read aloud: “ X at t -sub- n -plus-one minus X at t -sub- n equals A -sub- n , the drift term b - Δt , plus B -sub- n , the diffusion term σ - Δt - W -sub- n , plus a remainder R -sub- n .” where R_n collects higher-order Itô–Taylor corrections. The EM increment keeps exactly $A_n + B_n$, so the local error is R_n . By Itô–Taylor expansion,

$$R_n = \frac{1}{2} \sigma \partial_x \sigma ((\Delta W_n)^2 - \Delta t) + O((\Delta t)^{3/2}) + \text{terms of order } \Delta t \cdot \Delta W_n.$$

Read aloud: “ R -sub- n equals one-half sigma times partial- x sigma times the quantity Δt - W -sub- n squared minus Δt , plus big- O of Δt to the three-halves, plus terms of order Δt times Δt - W -sub- n .” The Brownian-increment scaling $\Delta W_n \sim \sqrt{\Delta t}$ implies:

- the pathwise (strong) local error has size $\sqrt{\Delta t} \cdot \Delta t = (\Delta t)^{3/2}$ per step from the $\Delta t \Delta W_n$ term, summed over $N = T/\Delta t$ steps and combined with Gronwall on the Lipschitz bound, this gives global strong error $\leq C \sqrt{\Delta t}$;
- the mean-zero terms $(\Delta W_n)^2 - \Delta t$ and ΔW_n cancel in expectation, so for test functions the leading error reduces by a half-power, yielding weak order 1.

A Grönwall argument on $e_n := \mathbb{E}|X_{t_n} - X_n|^2$, using the Lipschitz hypothesis on b, σ and the Itô isometry, then closes the recursion $e_{n+1} \leq (1 + C\Delta t) e_n + C(\Delta t)^2$, giving $e_n \leq C\Delta t$, hence $\mathbb{E}|X_{t_n} - X_n| \leq C \sqrt{\Delta t}$. $\square$

Remark 2 (Higher-order schemes). *Adding the Itô–Taylor term $\frac{1}{2} \sigma \partial_x \sigma ((\Delta W_n)^2 - \Delta t)$ to the increment gives the Milstein scheme, with strong order 1.*

3 Worked example: Ornstein–Uhlenbeck

For $dX_t = -\theta X_t dt + \sigma dW_t$, the integrating factor $e^{\theta t}$ gives

$$X_t = X_0 e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s, \quad \text{Var}(X_t) = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta t}) \xrightarrow{t \rightarrow \infty} \frac{\sigma^2}{2\theta}.$$

Read aloud: “ $X_{\text{sub-}t}$ equals X_{naught} times $e^{-\theta t}$, plus σ times the Itô integral from zero to t of $e^{-\theta(t-s)} dW_{\text{sub-}s}$; the variance of $X_{\text{sub-}t}$ equals σ^2 over 2θ times one minus $e^{-2\theta t}$, which tends to σ^2 over 2θ as t goes to infinity.” The companion code verifies the stationary variance empirically by averaging X_T^2 over many EM-simulated paths at large T .